

Bodhi McNally

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RESEARCH INTERESTS

Urological oncology; robotic and minimally invasive urological surgery; surgical outcomes; perioperative and functional outcomes; clinical prediction modelling; real-world clinical data; health services research.

EDUCATION

University of Sydney <i>Doctor of Medicine (MD)</i> Westmead Clinical School	Sydney, NSW 2026 – Expected Dec 2029
University of Sydney <i>Bachelor of Science (Data Science and Genetics & Genomics)</i> - Weighted Average Mark: 89.2 - Dalyell Scholars stream	Sydney, NSW 2022 – 2024

AWARDS & SCHOLARSHIPS

DKSH Inspire Award Recognised for exceptional performance, innovation and collaborative excellence while working across DKSH Healthcare pharmaceutical programs.	2025
Academic Merit Prize (x3) Awarded to students achieving outstanding academic performance across all undergraduate majors (approximately top 1% of students).	2022–2024
Dean's List for Academic Excellence (x3) Awarded to students achieving exceptional academic performance within the Faculty of Science (approximately top 3% of students).	2022–2024
Science Top of the State Scholarship Awarded for achieving the highest mark in a New South Wales HSC Science course. 🔗	2022
Dean's Entry Scholarship Competitive scholarship awarded to high-achieving commencing students. Awarded on the basis of academic excellence (ATAR 99.75), leadership potential and extracurricular achievement.	2022
Sydney Scholars Scholarship Merit scholarship awarded to outstanding commencing students demonstrating academic excellence, leadership potential and extracurricular achievement.	2022
Dalyell Scholar Invitation-only enrichment program for high-achieving students, providing access to advanced coursework, mentoring and interdisciplinary learning opportunities.	2022–2024
E12 Scholarship University of Sydney scholarship recognising academic achievement, leadership and community engagement among students from under-represented backgrounds.	2022

RESEARCH APPOINTMENTS & EXPERIENCE

Surgical Research Scholar <i>Research, Innovation & Surgical Education (RISE)</i> - Selected for the 2026 RISE Scholar cohort following a competitive application and panel interview process. - Participate in surgical research meetings, academic development activities and collaborative research opportunities with surgical mentors and student researchers. - Research interests include urological oncology, robotic and minimally invasive surgery, surgical outcomes, clinical prediction modelling and the use of real-world clinical data to evaluate and improve surgical care.	Sep 2026 – Present Sydney, NSW
Advanced Statistical Consultant <i>STAT3926/STAT4026 Collaborative Research Program</i> 🔗	2024 University of Sydney
- Selected to undertake advanced statistical consulting projects alongside Honours students enrolled in STAT4026, providing quantitative support to Higher Degree Research candidates and academic researchers. - Applied statistical modelling, model diagnostics and reproducible analytical workflows to interdisciplinary research questions using complex observational datasets.	
Multi-Omics Analysis of <i>Escherichia coli</i> Adaptation to Blood Environments 🔗 <i>Bachelor of Science Capstone Research Project</i>	2024 University of Sydney
- Investigated metabolic and proteomic adaptations of <i>Escherichia coli</i> strain B36 during growth in human blood compared with nutrient broth, with relevance to bloodstream infection and sepsis pathogenesis. - Performed differential protein-expression and STRINGdb network analyses, identifying enrichment of bacterial chemotaxis, flagellar biosynthesis and tricarboxylic acid cycle pathways among upregulated proteins. - Analysed LC-MS/MS metabolomic data comprising 95 metabolites; principal component analysis demonstrated separation between blood-grown and control cells, with pathway enrichment implicating glutathione metabolism, sulphur amino-acid biosynthesis and central carbon metabolism. - Integrated proteomic and metabolomic findings to characterise coordinated changes in chemotaxis, energy production and adaptive nutrient acquisition associated with host adaptation.	
Assessing Coral Recovery and Reassembly in the Great Barrier Reef 🔗 <i>Client: Dr Ana Paula da Silva, School of Geosciences</i>	2024 University of Sydney
Investigated ecological recovery following disturbance events across Great Barrier Reef reefs using long-term monitoring and environmental datasets.	

- Quantified community reassembly using Bray–Curtis similarity indices and modelled successful reassembly using logistic regression, finding that 83% of reefs recovered pre-disturbance community composition.
- Applied Wilcoxon rank-sum tests to compare recovery times across coral morphologies, regions and shelf locations, identifying substantial spatial and taxonomic variation in resilience.
- Developed generalised additive models to examine projected changes in disturbance frequency under RCP 4.5 and RCP 8.5 climate scenarios using sea-surface-temperature projections.

Statistical Consulting for Ecological Modelling Research

2024

Client: Mei-Ting Kao (PhD Candidate), School of Life and Environmental Sciences

University of Sydney

- Provided statistical consulting support for research investigating habitat-associated variation in *Chalinolobus gouldii* vocalisation behaviour.
- Evaluated and validated Poisson generalised linear mixed models incorporating nested random effects to account for hierarchical sampling across sites and recording periods.
- Performed model diagnostics using DHARMA, including assessment of overdispersion, zero inflation and residual behaviour.
- Identified associations between vocalisation activity and temperature, reproductive status and anthropogenic disturbance while quantifying site-level variation through random-effects modelling.

CRISPR-Cas9 Editing of CCR5: Sequence, Structure and Bioethical Analysis

2023

Genomics Research Project

University of Sydney

- Investigated the molecular and ethical implications of CRISPR-Cas9-mediated editing of the CCR5 gene in the context of human germline modification.
- Performed sequence alignment and protein-translation analyses of wild-type CCR5, the naturally occurring $\Delta 32$ variant and edited alleles generated in the He Jiankui experiment.
- Applied transmembrane-domain prediction and structural analysis to evaluate the functional consequences of frameshift mutations and altered receptor architecture.
- Critically evaluated gene-editing methodology, including PAM-site selection, DNA-repair mechanisms and off-target effects, alongside the ethical implications of heritable genome editing.

TEACHING & EDUCATIONAL SCHOLARSHIP

Detailed teaching portfolio available online: 

Tutor and Casual Academic

Feb 2024 – Present

School of Mathematics and Statistics

University of Sydney

- Tutor for DATA1001: Introduction to Data Science  and MATH1062: Mathematics 1B , large first-year units enrolling more than 2,000 students annually.
- Facilitate workshops of up to 60 students covering statistics, data science, calculus, differential equations and linear algebra.
- Develop and deliver interactive learning activities using real-world datasets to strengthen students' conceptual understanding and practical application of statistical, mathematical and programming principles.
- Assess major student projects with a focus on analytical rigour, reproducibility and scientific communication, providing detailed formative feedback across hundreds of student submissions.
- Mentor junior demonstrators in teaching practice, assessment standards and classroom management, supporting their progression into tutoring roles.
- Collaborate with academic staff to maintain consistency in assessment, marking standards, curriculum delivery and student support across large-enrolment units.

Curriculum Review and Development Team Member, DATA1x Program

Jul 2024 – Present

School of Mathematics and Statistics

University of Sydney

- Collaborate within a Students-as-Partners framework to review, innovate and redevelop curriculum across large-enrolment data science units serving thousands of students.
- Contribute to assessment redesign, curriculum innovation and learning-resource development to improve student engagement and learning outcomes.
- Develop best-practice marking guidelines to improve consistency, fairness and quality of assessment feedback.

Co-author, Global Students as Partners Roundtable 2026

May 2026 – Present

University of Sydney

Sydney, NSW

- Co-author of an educational scholarship submission entitled *The Dream Teams – Student-Led Curriculum Review, Innovation & Development in Massive Data Science Cohorts*.
- Exploring how Students-as-Partners models can support scalable curriculum innovation, assessment improvement and continuous quality enhancement across large undergraduate programs.

LEADERSHIP, GOVERNANCE & SERVICE

Member, MD Admissions Student Reference Group

Sydney Medical School

Jul 2026 – Present

University of Sydney

- Selected to contribute student perspectives to consultation and improvement initiatives relating to admissions to the Doctor of Medicine program.
- Participate in discussions with academic leadership concerning applicant experience, admissions processes and the ongoing development of medical student selection.

President

Sydney University Science Society

Oct 2023 – Dec 2024

University of Sydney

- Led Australia's largest undergraduate student society, representing more than 3,100 members and managing an executive team of 15 student leaders.
- Oversaw a \$176,000 operational budget, secured increased faculty funding and raised more than \$20,000 for equity initiatives that enabled 100+ students to attend high-cost events free of charge.
- Established the Society's first Bylaws, introducing governance frameworks for equity, environmental sustainability, privacy and financial management.
- Directed strategic growth initiatives including a marketing overhaul that increased social media reach by more than 3,000 followers and revitalised the Society's publication portfolio through the launch of its first First Year Guide in over 20 years.
- Secured strategic partnerships and mentored emerging student leaders, strengthening organisational sustainability and succession planning.

Student Member, Academic Board

University of Sydney

Jan 2023 – Jan 2024

Sydney, NSW

- Represent the Faculty of Science in university-level academic governance, contributing student perspectives to matters of academic policy, standards, quality assurance and educational strategy.
- Participate in deliberations concerning curriculum, assessment, student progression and the quality of teaching and learning across the University.

Student Representative, Faculty of Science Board

Faculty of Science

Jan 2023 – Jan 2024

University of Sydney

- Represented Science students in faculty-level governance and contributed to discussions concerning curriculum, assessment, academic standards, student experience and educational quality.
- Collaborated with academic leaders, professional staff and student representatives on faculty policy and strategic academic initiatives.

PROFESSIONAL EXPERIENCE

Data Analyst

AstraZeneca

Apr 2025 – Jan 2026

Macquarie Park, NSW

- Applied advanced analytical and data science methodologies to large-scale healthcare datasets to investigate patterns of disease burden, treatment utilisation and healthcare delivery across Australia and New Zealand.
- Designed and maintained reproducible data infrastructure and analytical workflows using SQL, R, Python, Amazon Redshift and Power BI, supporting both exploratory analyses and routine evidence generation.
- Integrated, validated and governed multiple real-world healthcare data sources, including prescribing, dispensing and market-level datasets, ensuring analytical robustness, data integrity and methodological consistency.
- Applied epidemiological and geospatial analytical techniques to characterise regional variation in disease burden, treatment access and healthcare utilisation across severe eosinophilic asthma, COPD, type 2 diabetes mellitus, heart failure and chronic kidney disease populations.
- Collaborated with multidisciplinary stakeholders to translate complex quantitative analyses into actionable insights through statistical reporting, data visualisation and evidence-based recommendations.

Key Account Manager – on behalf of AbbVie

DKSH Healthcare Australia

Jan 2025 – Apr 2025

Sydney, NSW

- Supported national engagement around AbbVie's curative hepatitis C treatment, working with healthcare professionals to strengthen patient recall, screening and treatment pathways.
- Increased healthcare-professional connect rate by 163%, expanding opportunities for clinical education, patient recall and prescribing engagement.
- Led recall-audit initiatives and trained field-team members in MedicalDirector and Best Practice software to support identification of patients potentially requiring hepatitis C follow-up.
- Used xBERT and Veeva CRM platforms to manage healthcare-professional interactions, monitor engagement and refine outreach and account prioritisation strategies.
- Generated and qualified more than 50 high-value leads for handover to the national field team, supporting targeted follow-up and prescribing opportunities.
- Built relationships with prescribers to support education, screening and patient identification across hepatitis C care pathways.

Program Coordinator – Partizan Health, on behalf of Novo Nordisk

DKSH Healthcare Australia

Jul 2024 – Dec 2024

North Sydney, NSW

- Coordinated patient-support activities across pharmaceutical programs, supporting continuity of care, patient engagement, pharmacovigilance and program compliance.

- Supported Novo Nordisk's WegovyCare program, managing patient interactions across telephone, email and live-chat channels and providing personalised program support to patients prescribed semaglutide for weight management.
- Maintained accurate pharmacovigilance documentation and escalation of reportable safety information in accordance with program and regulatory requirements.
- Recorded patient interactions in the Guardian CRM to maintain continuity, traceability and accurate program documentation.
- Supported Roche's Hemlibra program by verifying proof of dispensing with pharmacists to maintain the integrity of the nationally subsidised patient support program.

RESEARCH TRAINING, CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Anatomy of the Chest, Neck, Abdomen, and Pelvis 	Sep 2026
<i>Yale School of Medicine</i>	<i>Online</i>
• Completed comprehensive human anatomy coursework spanning the thorax, abdomen, retroperitoneum, pelvis and associated anatomical relationships.	
Accredited Urology Continuing Medical Education 	Sep 2026
<i>European Association of Urology / European School of Urology</i>	<i>Online</i>
• Completed 12 EACCME-accredited educational activities, each awarding 1 European CME credit (ECMEC), for a total of 12 ECMECs. • Through an agreement between the European Union of Medical Specialists (UEMS) and the American Medical Association (AMA), EACCME credits can be converted to an equivalent number of AMA PRA Category 1 Credits. ◦ <i>Mastering Excellence in Surgery – The Core Skills</i>  ◦ <i>EAU Guidelines on Urolithiasis</i>  ◦ <i>How to Manage Hypercalciuria in 2026</i>  ◦ <i>Antibiotic Prophylaxis in Urological Interventions</i>  ◦ <i>Update on Strategies for Diagnosis and Histological Characterisation of Renal Masses</i>  ◦ <i>Holistic Approach to Male LUTS</i>  ◦ <i>Update in the Management of Recurrent Urinary Tract Infection</i>  ◦ <i>EAU Guidelines on Non-Muscle Invasive Bladder Cancer</i>  ◦ <i>News in Testicular Masses Management in Clinical Practice</i>  ◦ <i>The Role of MRI in Prostate Cancer Screening and Diagnosis</i>  ◦ <i>The Role of Risk Stratification of Prostate Cancer</i>  ◦ <i>Key Updates in the 2025 EAU Urological Infections Guidelines</i> 	
ICH E6(R3) Good Clinical Practice 	Aug 2026
<i>Australian Clinical Trials Education Centre (A-CTEC)</i>	<i>Online</i>
• Completed Good Clinical Practice training aligned with the ICH E6(R3) guideline for the ethical and scientifically rigorous conduct of clinical research.	

QUANTITATIVE, COMPUTATIONAL & RESEARCH METHODS

Programming: R, Python, SQL
Statistical Methods: hypothesis testing; linear and logistic regression; generalised linear models; mixed-effects modelling; generalised additive models; time-series analysis and forecasting
Machine Learning: classification; clustering; dimensionality reduction; segmentation modelling; XGBoost; scikit-learn; TensorFlow
Data Analysis & Scientific Computing: pandas; NumPy; tidyverse; dplyr; fable
Data Management & Querying: PostgreSQL; Amazon Redshift; SQLite; DBever; Power Query; relational data manipulation; reproducible analytical workflows
Data Visualisation & Reporting: ggplot2; Matplotlib; Seaborn; Power BI; DAX
Data Engineering & Automation: Power Automate; PowerApps; Excel Macros
Bioinformatics & Genomics: BLAST; UniProt; STRINGdb; PLINK; IGV; VCF/GTF data formats
Geospatial Analysis: QGIS; ArcGIS Pro; leaflet; sf; geopandas; raster
Scientific Writing & Reproducibility: Git; Quarto; R Markdown; $\LaTeX$; Reveal.js

PROFESSIONAL MEMBERSHIPS

Australasian Students' Surgical Association (ASSA)	2026 – Present
Australian Medical Students' Association (AMSA)	2026 – Present
Sydney University Surgical Society	2026 – Present
Sydney University Medical Society	2026 – Present
Royal Australasian College of Surgeons (RACS) – Medical Student Member	2026 – Present
American Urological Association (AUA) – Medical Student Member	2026 – Present
European Association of Urology (EAU) – Student Member	2026 – Present
Australian Medical Association (NSW) – Student Member	2026 – Present

Referees available upon request.